Software requirement specification document for project InclusiveHire

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1 Introduction

1.1 Project Mission Statement

The mission of the InclusiveHire Platform is to create an accessible and intelligent recruitment system that connects candidates with disabilities to inclusive employers. The platform aims to reduce employment barriers by providing accessible tools, AI-powered job matching, and assistive technologies that enable individuals with different abilities to search for jobs, apply for opportunities, and communicate with employers effectively.

By integrating technologies such as OCR, natural language processing, and speech processing APIs, the platform promotes inclusive hiring practices and helps organizations identify qualified candidates based on their skills and experience rather than their limitations.

1.2 Purpose of this document

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the InclusiveHire Platform, including its functionality, system architecture, and operational requirements. This document defines the system's features, constraints, and interfaces to guide the development process and ensure that the final product meets the needs of both candidates with disabilities and employers.

This document serves as a reference for all stakeholders involved in the project, including software developers, system designers, project managers, testers, and academic supervisors. It outlines the system's functional and non-functional requirements, describes the system components and their interactions, and establishes a clear framework for the implementation and evaluation of the platform.

By documenting these requirements, the SRS ensures that the development team maintains a consistent understanding of the system's objectives and provides a structured approach for building an accessible and intelligent recruitment platform.

1.3 Scope of this document

This Software Requirements Specification (SRS) document defines the requirements for the InclusiveHire Platform, an accessible recruitment system designed to connect employers with candidates with disabilities. The document specifies the functional and non-functional requirements of the system, the system architecture, and the technologies involved in building the platform. It serves as a comprehensive reference for the development, implementation, and evaluation of the system.

The scope of this requirements definition effort focuses on designing and documenting a web-based platform that provides two primary portals: a Candidate Portal and an Employer Portal. The Candidate Portal enables individuals with disabilities to register, create profiles, upload CVs, and apply for job opportunities through an accessible interface. The Employer Portal allows organizations to post job openings, define job requirements, and identify suitable candidates through an AI-powered compatibility matching system. The platform also integrates several advanced technologies, including Optical Character Recognition (OCR) for CV data extraction, Natural Language Processing (NLP) for skill analysis, and speech processing APIs that convert text to speech and speech to text to support users with visual or hearing impairments.

The requirements elicitation process involved contributions from several key stakeholders. These include end users, such as individuals with disabilities who will interact with the candidate portal; employers and recruiters, who will use the employer portal to publish job opportunities and evaluate applicants; system engineers and designers, responsible for defining the system architecture and technology stack; and software developers, who will implement and maintain the platform. Academic supervisors and project advisors may also participate in reviewing the requirements to ensure the system meets the expected technical and academic standards.

Several constraints were considered during the requirements elicitation process. The development of the platform is limited by the project timeline, which requires the system to be implemented within a defined academic schedule. Additionally, the project operates under cost constraints, meaning that preference will be given to open-source tools and publicly available APIs where possible. The system will be developed within a standard software engineering environment, using modern web development technologies such as frontend frameworks (e.g., React or Vue), backend technologies (e.g., Node.js, Django, or PHP), and a relational database system. Integration with external services such as OCR engines and speech processing APIs may also depend on the availability and usage limits of these services.

By defining the scope and constraints of the project, this document ensures that all stakeholders have a clear understanding of the system objectives, the development environment, and the boundaries within which the InclusiveHire Platform will be designed and implemented.

1.4 Overview

The product defined in this document is the InclusiveHire Platform, an accessible web-based recruitment system designed to support employment opportunities for individuals with disabilities. The platform aims to bridge the gap between job seekers with disabilities and organizations looking to hire qualified candidates while ensuring accessibility and inclusivity throughout the recruitment process.

The system consists of two main portals: the Candidate Portal and the Employer Portal. The Candidate Portal allows individuals with disabilities to create profiles, upload CVs, and apply for job opportunities through an accessible interface designed to accommodate different user needs. The Employer Portal enables companies to post job openings, specify job requirements, and identify suitable candidates using an intelligent compatibility matching system.

To enhance functionality and accessibility, the platform integrates several advanced technologies. These include Optical Character Recognition (OCR) to extract relevant information from uploaded CVs, Natural Language Processing (NLP) to analyze candidate skills and experience, and speech processing APIs that convert text to speech and speech to text to support visually impaired and hearing-impaired users. The system also incorporates accessibility features such as high-contrast display modes, screen reader compatibility, keyboard navigation, and subtitle support for multimedia content.

Overall, the InclusiveHire Platform provides an intelligent and accessible recruitment environment that improves the job search experience for candidates with disabilities while helping employers efficiently identify suitable applicants based on their abilities and job requirements.

1.5 Business Context

The development of the InclusiveHire Platform is driven by the need to promote equal employment opportunities and improve accessibility within the recruitment process. Many individuals with disabilities face challenges when searching for jobs due to limited accessibility in traditional recruitment platforms and the lack of tools that help employers understand how to accommodate different abilities. This project aims to address these challenges by providing a technology-driven platform that supports inclusive hiring practices.

The mission of the organization sponsoring the development of this platform is to empower individuals with disabilities by providing accessible digital tools that connect them with employment opportunities and supportive workplaces. By leveraging modern technologies such as artificial intelligence, speech processing APIs, and optical character recognition, the platform aims to reduce barriers in the job application process and create a more inclusive employment ecosystem.

The primary organizational objectives of the project include improving accessibility in online recruitment systems, supporting employers in identifying suitable candidates based on their abilities rather than their limitations, and encouraging inclusive workplace practices. The platform also seeks to demonstrate how advanced technologies can be used to support individuals with disabilities in professional environments. In addition, the system aims to promote awareness of inclusive hiring while providing employers with tools that help them make fair and informed hiring decisions.

Through these objectives, the InclusiveHire Platform supports the broader goal of building a more inclusive labor market where individuals with disabilities have equal opportunities to contribute their skills and talents.

2 General Description

2.1 Product Functions

Describes the general functionality of the product, which will be discussed in more detail below.

2.2 Similar System Information

The InclusiveHire Platform is a standalone web-based recruitment system designed specifically to connect people with disabilities to inclusive employers using AI-powered matching and accessibility-first design principles.

Existing recruitment platforms such as LinkedIn and Indeed provide general job search and recruitment services. However, these platforms are not specifically designed to address accessibility challenges or evaluate physical and ability-based job compatibility.

Organizations like Specialisterne focus on inclusive employment initiatives, but they do not provide a fully integrated AI-driven compatibility scoring system combined with OCR-based CV analysis and built-in accessibility APIs.

The InclusiveHire Platform differs by:

- Providing AI-based ability-focused job matching
- Generating compatibility scores (0–100%)
- Supporting accessibility features (text-to-speech, speech-to-text, high-contrast mode, ARIA labels)
- Allowing employers to define physical requirements and accommodation needs
- Offering automated accommodation suggestions

The system is designed as a standalone product but may be integrated in the future with government disability verification systems or external HR management platforms.

2.3 User Characteristics

The system serves three primary user groups:

2.3.1 Candidates (People with Disabilities)

- Basic to intermediate computer literacy
- May depend on assistive technologies (screen readers, voice navigation)
- May have visual, hearing, mobility, or cognitive impairments
- Require a simple and accessible user interface

2.3.2 Employers / HR Representatives

- Intermediate to advanced computer literacy
- Familiar with recruitment systems
- Basic understanding of inclusive hiring practices
- Responsible for defining job requirements and workplace accommodations

2.3.3 System Administrators

- Advanced technical expertise
- Responsible for system maintenance, monitoring AI components, and managing user accounts

The platform must prioritize usability, clarity, and accessibility due to the diverse abilities of its users.

2.4 User Problem Statement

People with disabilities face significant challenges in accessing employment opportunities. Current recruitment platforms:

- Do not assess physical or ability-based compatibility
- Lack built-in accessibility support
- Do not provide accommodation recommendations
- Depend heavily on manual CV screening

Employers also experience difficulties such as:

- Identifying suitable candidates based on ability compatibility
- Structuring inclusive job descriptions
- Reducing unconscious bias during hiring

As a result, there is a gap between qualified candidates with disabilities and inclusive employers. A specialized system is needed to bridge this gap using intelligent matching and accessibility-driven design.

2.5 User Objectives

From the users' perspective, the system should achieve the following objectives:

2.5.1 Candidate Objectives

- Register through an accessible interface
- Select disability type and ability profile
- Upload CV and receive an AI-generated summary
- Receive job recommendations based on compatibility scoring
- Obtain accommodation suggestions

2.5.2 Employer Objectives

- Post detailed job requirements
- Define physical and environmental needs
- View AI-generated compatibility scores (0–100%)
- Access a dashboard of matched candidates
- Promote inclusive hiring practices

2.5.3 Desired (Future) Features

- Government disability verification integration
- AI-powered mock interviews
- Sign language avatar support
- Inclusive employer certification badges

The overall objective is to create a fair, intelligent, and accessible recruitment ecosystem that supports inclusive employment.

2.6 General Constraints

2.6.1 Technical Constraints

- Web-based system (React/Vue frontend, Node.js/Django/PHP backend)
- Integration with OCR tools (e.g., Tesseract or Google Vision API)
- Integration of NLP modules for skill extraction
- Support for accessibility APIs (text-to-speech and speech-to-text)

2.6.2 Performance Constraints

- Compatibility scoring response time $\leq$ 3 seconds
- CV processing time $\leq$ 5 seconds
- System availability $\geq$ 99% uptime

2.6.3 Hardware Constraints

- Must operate on standard web browsers
- Must be compatible with assistive technologies (screen readers, voice control systems)

2.6.4 Compliance Constraints

- Must follow WCAG accessibility guidelines
- Must ensure secure handling of user data
- Must comply with relevant data protection regulations

2.7 System Boundary and External Entities

The InclusiveHire platform operates as an online recruitment system designed to connect candidates with disabilities to employers. The system boundary defines the internal components that belong to the platform and the external entities that interact with it.

2.7.1 Internal System Components

The following components are considered part of the InclusiveHire platform:

- Candidate Portal: allows candidates to register, upload CVs, search for jobs, and apply for opportunities.
- Employer Portal: enables employers to post job openings and review candidate applications.
- Matching Engine: compares candidate skills with job requirements and generates compatibility scores.
- CV Processing Module: analyzes uploaded CVs using OCR and text analysis techniques.
- Accessibility Module: provides accessibility tools such as text-to-speech, speech-to-text, keyboard navigation, and high-contrast mode.
- Notification System: sends notifications to candidates and employers regarding applications and job opportunities.

2.7.2 External Entities

The system interacts with the following external entities:

- Candidates with Disabilities: users who search for jobs and apply for opportunities.
- Employers: companies that post jobs and review applications.

- OCR Services: external OCR engines used to extract text from uploaded CV files.
- Speech Processing APIs: external services that support text-to-speech and speech-to-text functionality.
- Email Notification Services: used to send system notifications.
- Cloud Infrastructure: hosting platforms that provide servers and storage resources.

2.8 Stakeholder Identification and Ranking

The InclusiveHire platform involves several stakeholders who influence the system requirements and development process. The following table presents the key stakeholders and their priority levels.

Table 1: Stakeholder Identification and Ranking

Stakeholder	**Description**	**Priority**
Candidates with Disabilities	Primary users who use the system to search for jobs, upload CVs, and apply for job opportunities.	Very High
Employers / HR Representatives	Organizations that post job openings and evaluate applicants.	Very High
System Administrators	Responsible for managing the system and ensuring smooth operation.	High
Accessibility Specialists	Experts who ensure that the platform follows accessibility standards.	Medium
Government / Regulatory Bodies	Organizations that may enforce employment and accessibility regulations.	Medium
Development Team	Responsible for building and maintaining the system.	Medium

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2.9 Rich Picture Diagram

A rich picture diagram is used to illustrate the overall structure of the InclusiveHire platform and the interactions between the main actors and system components.

The diagram shows how candidates with disabilities interact with the candidate portal to upload CVs and search for jobs, while employers interact with the employer portal to publish job opportunities and review applicants. The system processes uploaded CVs using OCR technology and analyzes candidate skills to match them with suitable jobs. Accessibility services such as text-to-speech and speech-to-text ensure that the platform remains usable for individuals with different types of disabilities.

External services such as OCR engines, speech processing APIs, and cloud infrastructure support the system's functionality.

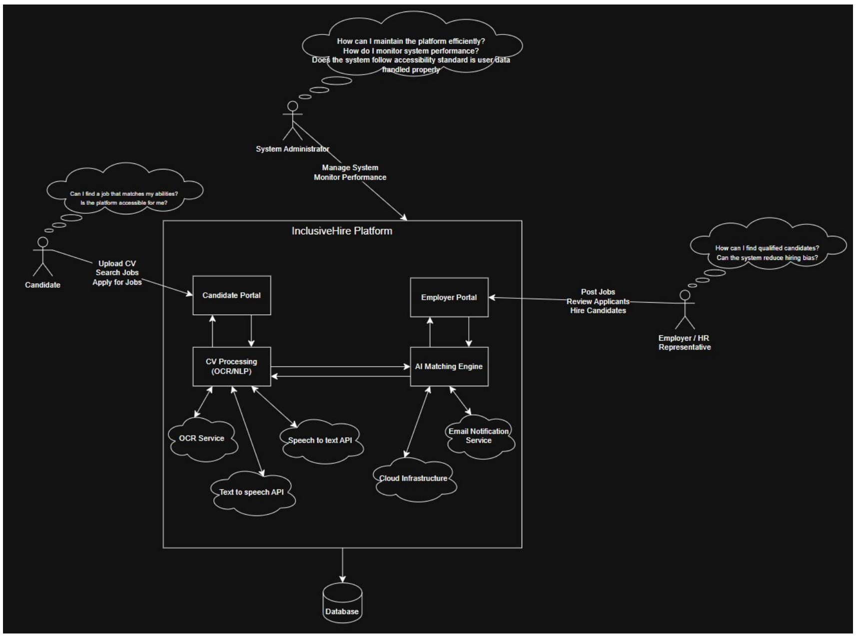

Figure 1: Enter Caption

3 Functional Requirements

This section lists the functional requirements in ranked order. Functional requirements describes the possible effects of a software system, in other words, what the system must accomplish. Other kinds of requirements (such as interface requirements, performance requirements, or reliability requirements) describe how the system accomplishes its functional requirements see table 11. Each functional requirement should be specified in a format similar to the following:

Table 2: Upload CV Files

Function Name	Allow candidates to upload CV files (PDF or DOC).
Description	The system shall allow registered candidates to upload their CV files in PDF or DOC format. The file will be stored securely in the system database or cloud storage.
Critically	High – Core functionality required for job applications.
Technical issues	File validation, file size limits, secure storage, and malware scanning must be implemented.
Cost and schedule	Medium – Requires backend storage configuration and validation logic.
Risks	Unsupported file formats or corrupted files may cause processing errors. File validation will reduce this risk.
Dependencies with other requirements	Depends on user authentication (registration/login). Required before OCR processing (FR19).
Pre-Condition	Candidate must be logged in.
Post-Condition	CV file is successfully uploaded and stored in the system.

Table 3: OCR Text Extraction

Function Name	Extract text from uploaded CV using OCR.
Description	The system shall process uploaded CV files using OCR technology to extract textual information such as name, skills, education, and work experience.
Critically	High – Required for automated skill analysis.
Technical issues	OCR accuracy, handling different CV formats, integration with OCR libraries.
Cost and schedule	Medium to High – Requires third-party OCR integration and testing.
Risks	Low OCR accuracy may extract incorrect data. Risk reduced by text validation and manual editing option.
Dependencies with other requirements	Depends on CV upload (FR18). Required before skill analysis (FR20).
Pre-Condition	A CV file must be uploaded successfully.
Post-Condition	Text content is extracted and stored for analysis.

Table 4: CV Skill Analysis

Function Name	Analyze extracted CV text to identify skills and experience.
Description	The system shall analyze the extracted CV text to detect and categorize candidate skills, qualifications, and years of experience.
Critically	High – Improves job matching accuracy.
Technical issues	NLP processing, keyword matching, skill database mapping.
Cost and schedule	Medium – Requires implementation of text parsing logic.
Risks	Incorrect skill detection due to formatting or uncommon terminology.
Dependencies with other requirements	Depends on OCR extraction (FR19).
Pre-Condition	Extracted text must be available.
Post-Condition	Candidate skills are identified and stored in profile.

Table 5: Text-to-Speech Accessibility

Function Name	Provide text-to-speech functionality.
Description	The system shall convert on-screen text into audio output to assist visually impaired users.
Critically	High – Essential accessibility feature.
Technical issues	Integration with browser TTS APIs, multilingual support.
Cost and schedule	Medium – Requires frontend integration.
Risks	Browser compatibility limitations.
Dependencies with other requirements	Depends on user interface module.
Pre-Condition	User enables accessibility mode.
Post-Condition	Text content is read aloud to the user.

Table 6: Keyboard Navigation Support

Function Name	Support full keyboard navigation.
Description	The system shall allow users to navigate all interface elements using only the keyboard.
Critically	High – Required for accessibility compliance.
Technical issues	Proper tab indexing, focus indicators, ARIA roles.
Cost and schedule	Low to Medium – Mainly frontend configuration.
Risks	Poor implementation may reduce usability.
Dependencies with other requirements	Depends on UI design.
Pre-Condition	User accesses the web interface.
Post-Condition	User can operate system without mouse.

Table 7: Employer Notification

Function Name	Notify employers when a candidate applies.
Description	The system shall send a notification (email or in-system alert) to employers when a candidate submits a job application.
Critically	High – Important for communication workflow.
Technical issues	Email service integration, real-time notification handling.
Cost and schedule	Medium – Requires backend event handling.
Risks	Email delivery failure.
Dependencies with other requirements	Depends on job application module.
Pre-Condition	Candidate submits job application.
Post-Condition	Employer receives notification.

Table 8: Speech-to-Text Accessibility

Function Name	Provide speech-to-text functionality.
Description	The system shall allow users to convert spoken input into text to assist users who have difficulty typing or hearing impairments. The converted text can be used in forms, search fields, and messaging features.
Critically	High – Important accessibility feature for inclusive interaction.
Technical issues	Integration with browser speech recognition APIs, language support, background noise handling.
Cost and schedule	Medium – Requires frontend API integration and testing.
Risks	Low speech recognition accuracy in noisy environments. Risk reduced by allowing manual text editing.
Dependencies with other requirements	Depends on user interface components and messaging module.
Pre-Condition	User enables speech input feature and grants microphone permission.
Post-Condition	Spoken words are converted into editable text in the input field.

Table 9: High-Contrast Display Mode

Function Name	Provide high-contrast display mode.
Description	The system shall provide a high-contrast visual mode to improve readability for users with visual impairments. This mode will adjust colors, font weight, and background contrast.
Critically	High – Required for accessibility compliance.
Technical issues	CSS theme switching, maintaining consistency across components, accessibility testing.
Cost and schedule	Low to Medium – Primarily frontend implementation.
Risks	Poor color combination may reduce usability. Testing with accessibility standards reduces risk.
Dependencies with other requirements	Depends on UI design module.
Pre-Condition	User activates high-contrast mode from settings.
Post-Condition	Interface switches to high-contrast theme.

Table 10: Job Match Notifications

Function Name	Notify candidates about matching job opportunities.
Description	The system shall automatically notify candidates when new job postings match their profile skills, experience, or accessibility preferences. Notifications may be sent via email or in-system alerts.
Critically	High – Enhances engagement and improves job matching efficiency.
Technical issues	Matching algorithm implementation, real-time notification service, email integration.
Cost and schedule	Medium – Requires backend matching logic and notification service.
Risks	Incorrect matching due to incomplete profile data. Risk reduced by improving skill extraction (FR20).
Dependencies with other requirements	Depends on skill analysis (FR20) and job posting module.
Pre-Condition	Candidate profile and job postings must exist in the system.
Post-Condition	Candidate receives notification about relevant job opportunity.

4 Interface Requirements

The InclusiveHire Platform interacts with different types of users and external software services to provide its functionality. These interfaces ensure that the system can communicate effectively with users, databases, and third-party technologies used for accessibility and artificial intelligence features.

4.1 User Interfaces

The system provides a web-based user interface that allows candidates and employers to interact with the platform through a standard web browser. The interface will include two main portals: the Candidate Portal and the Employer Portal.

The Candidate Portal enables users with disabilities to register, upload CVs, search for jobs, and apply for job opportunities through an accessible interface. The Employer Portal allows companies to post job openings, define job requirements, and review candidate applications.

4.1.1 GUI

The InclusiveHire Platform provides a web-based graphical user interface that allows both candidates and employers to interact with the system through standard web browsers. The interface is designed to be simple, accessible, and user-friendly, ensuring that individuals with disabilities can easily navigate and use the platform.

The system contains two primary graphical portals: the Candidate Portal and the Employer Portal. The Candidate Portal allows users to register, create profiles, upload CVs, view job recommendations, and apply for jobs. The Employer Portal

Table 11: Candidate Registration

Function Name	Candidate Registration
Description	Allows candidates with disabilities to create an accessible account on the platform
Critically	High – required for users to access system services.
Technical issues	Must support secure authentication and accessibility standards
Cost and schedule	DLow–Medium development effort during initial development phase.
Risks	D Incorrect user data entry or accessibility compatibility issues.
Dependencies with other requirements	Depends on authentication and user database modules.
Pre-Condition	User has internet access and opens the registration page.
Post-Condition	A new candidate account is created and stored in the database.

enables companies to create accounts, post job opportunities, define job requirements, and review candidate applications.

The graphical interface follows accessibility guidelines by including features such as high-contrast display mode, keyboard navigation support, screen reader compatibility through ARIA labels, and clear visual layouts. These features ensure that users with visual or physical impairments can interact with the platform effectively.

The main interface screens of the system include:

Registration and Login Screen: Allows candidates and employers to create accounts and access the platform.

Candidate Dashboard: Displays profile information, job recommendations, and application status.

Employer Dashboard: Displays job postings, candidate matches, and application management tools.

Job Posting Interface: Allows employers to define job requirements and publish job opportunities.

CV Upload Interface: Allows candidates to upload CV files, which are processed using OCR technology.

These interface screens can be represented using mock-ups or screenshots in the final system documentation to illustrate how users interact with the platform.

4.1.2 CLI

The InclusiveHire Platform is primarily designed as a web-based application with a graphical user interface, and therefore it does not require a command-line interface for regular users. All core functionalities such as user registration, job posting, CV upload, and job matching are performed through the graphical web interface.

However, a limited command-line interface may be used by system administrators or developers for system maintenance and management tasks. These tasks may include database management, system configuration, and running background services such as OCR processing or AI model updates.

4.1.3 API

The InclusiveHire Platform uses a set of RESTful APIs to enable communication between the frontend interface, backend services, and external systems such as OCR engines and speech processing services. These APIs allow the system to process user requests, manage job postings, analyze CVs, and support accessibility features.

The APIs are accessed through HTTP/HTTPS requests, and data is exchanged in JSON format. Authentication for protected API endpoints is handled through secure authentication mechanisms such as session tokens or JSON Web Tokens (JWT).

4.1.4 Diagnostics or ROM

The InclusiveHire Platform provides diagnostic and debugging mechanisms to assist developers and system administrators in monitoring system performance and identifying potential errors. Diagnostic information may include system logs, error messages, and API response logs generated by the backend services.

The system records important events such as user authentication attempts, CV processing operations, job posting activities, and API interactions. These logs are stored in the server environment and can be accessed by authorized administrators for troubleshooting and system maintenance.

Additionally, the platform may include debugging tools during development, such as server console logs and database monitoring utilities, which help developers identify issues related to data processing, OCR extraction, AI matching algorithms, and external API integrations.

These diagnostic features ensure that system errors can be detected, analyzed, and resolved efficiently while maintaining system reliability and performance.

4.2 Hardware Interfaces

The InclusiveHire Platform is a web-based system that operates on standard computing devices and does not require specialized hardware components. Users can access the platform using common devices such as desktop computers, laptops, tablets, and smartphones through a web browser.

The system interacts with standard input and output hardware devices, including keyboards, mice, microphones, speakers, and display screens. These devices enable users to interact with the system interface, upload documents, and use accessibility features such as speech-to-text and text-to-speech services.

For accessibility purposes, the system may also support assistive hardware technologies such as screen readers, voice input devices, and audio output devices that help users with visual or hearing impairments interact with the platform effectively.

Overall, the system relies on commonly available hardware devices to ensure compatibility and accessibility for a wide range of users.

4.3 Communications Interfaces

The InclusiveHire Platform communicates with external systems and services through standard network communication protocols over the internet. The system uses HTTP and HTTPS protocols to enable secure communication between the client-side web interface and the backend server.

Data exchanged between the system components and external services is formatted using JSON (JavaScript Object Notation) through RESTful API requests. These communication interfaces allow the system to interact with external technologies such as OCR services for CV processing, AI matching services, and speech processing APIs for accessibility features.

Additionally, the platform supports secure data transmission using encryption protocols to protect sensitive user information such as personal details, login credentials, and job application data. This communication infrastructure ensures reliable interaction between users, the web application, and integrated third-party services.

4.4 Software Interfaces

The InclusiveHire Platform interacts with several external software systems and services to support its core functionality. These software interfaces enable the system to process user data, analyze uploaded documents, and provide accessibility features for users with disabilities.

The system integrates with Optical Character Recognition (OCR) software to extract text and relevant information from uploaded CV documents. This interface allows the platform to automatically analyze resumes and populate candidate profiles with extracted skills, education, and work experience.

Additionally, the platform integrates with Artificial Intelligence and Natural Language Processing (NLP) services to analyze candidate qualifications and generate compatibility scores between candidates and job postings. These services assist in recommending suitable job opportunities for candidates.

The system also communicates with speech processing APIs that provide text-to-speech and speech-to-text functionality. These interfaces support accessibility features for visually impaired and hearing-impaired users by converting written text into audio and spoken input into text.

All software interfaces communicate with the platform through secure API connections, typically using RESTful services and data formats such as JSON to ensure efficient and reliable data exchange between the system and external services.

5 Performance Requirements

The InclusiveHire Platform must meet certain performance requirements to ensure that the system operates efficiently and provides a responsive user experience. The system should be capable of handling multiple users simultaneously without significant delays in processing requests.

The platform should respond to standard user actions, such as page navigation, job searches, and profile updates, within 2–3 seconds under normal operating conditions. Processes that require additional computation, such as CV analysis using OCR or AI-based job matching, may take slightly longer but should typically complete within 5–10 seconds.

In terms of memory and resource usage, the system should be optimized to operate efficiently on standard server hardware. The backend server should manage memory usage effectively to support concurrent user sessions and API integrations. The system should also ensure that database queries and API requests are processed efficiently to minimize latency.

These performance requirements ensure that the platform remains responsive, scalable, and capable of supporting a growing number of users while maintaining reliable system performance.

6 Design Constraints

The development of the InclusiveHire Platform must follow several design constraints that influence how the system is implemented. These constraints ensure that the system meets technical, accessibility, and project requirements.

First, the system must comply with web accessibility standards to support users with disabilities. This includes compatibility with screen readers, keyboard navigation, high-contrast display modes, and integration with speech technologies such as text-to-speech and speech-to-text.

Second, the platform must be developed using web-based technologies that allow users to access the system through standard web browsers without requiring specialized software installation. The system architecture should support integration with external services such as OCR tools for CV processing and AI models for job matching.

Another constraint is the project timeline and resource availability, as the system is being developed within a limited academic schedule. Therefore, the development team may rely on existing APIs and open-source tools instead of building all components from scratch.

In addition, the system must follow data security and privacy requirements, ensuring that sensitive user information such as personal details, CV data, and job applications is protected through secure authentication mechanisms and encrypted communication protocols.

These constraints guide the design team in building a system that is accessible, secure, and feasible within the available technical and time resources.

6.1 Standards Compliance

The InclusiveHire Platform shall comply with recognized international standards and best practices to ensure accessibility, security, and interoperability.

- The system shall comply with the Web Content Accessibility Guidelines (WCAG) 2.1 Level AA to ensure accessibility for users with disabilities.
- The system shall follow ISO/IEC 27001 principles for information security management.
- The system shall comply with applicable data protection and privacy regulations.
- The system shall follow RESTful API standards for integration with external services.
- The system shall adhere to secure coding standards and industry best practices.

Compliance with these standards ensures the platform provides an inclusive, secure, and reliable environment for all users.

6.2 Hardware Limitations

The system is designed as a web-based application and must operate within the following hardware constraints:

- The platform shall run on standard desktop and laptop computers with modern web browsers (Chrome, Firefox, Edge, Safari).
- The system shall be compatible with assistive hardware devices such as screen readers, alternative input devices, and voice control systems.
- The system shall support mobile devices with standard specifications.
- Server infrastructure must support scalable cloud hosting environments.

No specialized hardware shall be required for end users beyond standard internet-enabled devices.

6.3 others as appropriate

6.3.1 Security Requirements

- The system shall ensure secure user authentication using encrypted credentials.
- The system shall implement role-based access control (RBAC) to restrict unauthorized actions.
- All data transmission shall be encrypted using HTTPS (TLS protocol).
- The system shall protect against common security threats such as SQL injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF).
- Regular security audits and vulnerability assessments shall be conducted.

6.3.2 Data Privacy Requirements

- The system shall collect only necessary user data.
- Users shall have the ability to update or delete their personal information.
- The system shall provide a clear privacy policy.
- Sensitive information (e.g., disability type) shall be securely stored and accessible only to authorized users.

6.3.3 Usability and Accessibility Requirements

- The user interface shall be simple, intuitive, and easy to navigate.
- The system shall support keyboard-only navigation.
- The platform shall provide high-contrast mode and adjustable font sizes.
- Text-to-speech and speech-to-text features shall be supported.

6.3.4 Maintainability Requirements

- The system architecture shall follow modular design principles.
- Source code shall be well documented.
- The system shall allow future enhancements without major restructuring.

6.3.5 Scalability Requirements

- The system shall support increasing numbers of users without significant performance degradation.
- The system shall allow deployment in scalable cloud environments.

Table 13: NON - Maintainability

Attribute Name	Maintainability
Description	The system shall be modular and well-documented to allow easy updates and bug fixes.
Critically	Medium – Reduces long-term maintenance cost.
Technical issues	Code documentation, modular design, version control usage.
Cost and schedule	Low to Medium – Requires development standards and documentation.
Risks	Poor documentation may increase debugging time.
Dependencies with other requirements	Affects all functional requirements.
Pre-Condition	Development standards are defined.
Post-Condition	Future modifications can be implemented efficiently.

Table 12: NON Reliability

Attribute Name	Reliability
Description	The system shall operate continuously with minimal downtime and handle unexpected errors gracefully.
Critically	High – Ensures system stability and user trust.
Technical issues	Error handling, exception management, database backup, and recovery mechanisms.
Cost and schedule	Medium – Requires testing and backup solutions.
Risks	System crashes or data loss if proper recovery mechanisms are not implemented.
Dependencies with other requirements	Depends on database integrity and server stability.
Pre-Condition	System infrastructure is properly configured.
Post-Condition	System remains operational even in case of minor failures.

Table 14: NON - Portability

Attribute Name	Portability
Description	The system shall be accessible on different browsers and operating systems.
Critically	Medium – Expands user accessibility.
Technical issues	Cross-browser compatibility testing and responsive design.
Cost and schedule	Medium – Requires UI and compatibility testing.
Risks	UI layout issues on different devices or browsers.
Dependencies with other requirements	Depends on frontend implementation.
Pre-Condition	System is developed using standard web technologies.
Post-Condition	Users can access the system from various platforms.

Table 15: NON - Performance (Resource Utilization)

Attribute Name	Performance / Resource Utilization
Description	The system shall respond to user requests within 3 seconds under normal load conditions.
Critically	High – Impacts user experience.
Technical issues	Database indexing, server optimization, efficient queries.
Cost and schedule	Medium – Requires performance testing.
Risks	Slow response time under heavy load.
Dependencies with other requirements	Depends on server infrastructure and database design.
Pre-Condition	Server resources are properly configured.
Post-Condition	System maintains acceptable response time.

Table 16: NON - Binary Compatibility

Attribute Name	Binary Compatibility
Description	The system shall maintain compatibility with existing compiled components, libraries, and APIs without requiring recompilation when minor updates occur.
Critically	Medium – Ensures smooth system upgrades and integration stability.
Technical issues	Version control management, backward compatibility handling, dependency management.
Cost and schedule	Medium – Requires careful release planning and regression testing.
Risks	Breaking changes may cause system modules to fail after updates.
Dependencies with other requirements	Depends on system architecture and external libraries.
Pre-Condition	Stable API and module interfaces are defined.
Post-Condition	Updates do not break existing integrations or components.

Table 17: NON - Extensibility

Attribute Name	Extensibility
Description	The system shall allow adding new features or modules without major redesign of existing components.
Critically	High – Supports future growth of the system.
Technical issues	Modular architecture, use of design patterns, API-based integrations.
Cost and schedule	Medium – Requires proper architectural planning.
Risks	Poor design may make future extensions complex and costly.
Dependencies with other requirements	Closely related to maintainability and system architecture.
Pre-Condition	System follows modular and layered design principles.
Post-Condition	New features can be integrated with minimal impact on existing functionality.

Table 18: NON - Reusability

Attribute Name	Reusability
Description	The system components shall be designed in a way that allows reuse in other applications or future projects.
Critically	Medium – Reduces development time for future systems.
Technical issues	Proper abstraction, separation of concerns, reusable libraries and services.
Cost and schedule	Low to Medium – Requires clean coding practices.
Risks	Tight coupling may limit reuse of components.
Dependencies with other requirements	Depends on maintainability and extensibility.
Pre-Condition	Components are implemented as independent modules.
Post-Condition	Modules can be reused in other systems with minimal modification.

Table 19: NON - Application Affinity / Compatibility

Attribute Name	Application Compatibility
Description	The system shall integrate and operate correctly with external applications such as email services, payment gateways, and third-party APIs.
Critically	High – Required for external service integration.
Technical issues	API integration, data format compatibility (JSON/XML), authentication tokens.
Cost and schedule	Medium – Requires integration testing.
Risks	Third-party API changes may disrupt system functionality.
Dependencies with other requirements	Depends on security and network configuration.
Pre-Condition	External services APIs are available and accessible.
Post-Condition	System successfully communicates with integrated applications.

Table 20: NON - Serviceability

Attribute Name	Serviceability
Description	The system shall provide logging, monitoring, and diagnostic tools to facilitate troubleshooting and support.
Critically	Medium – Assists in faster issue resolution.
Technical issues	Error logging, monitoring dashboards, system alerts.
Cost and schedule	Medium – Requires monitoring tool integration.
Risks	Lack of proper logs may increase downtime during failures.
Dependencies with other requirements	Depends on reliability and infrastructure setup.
Pre-Condition	Logging and monitoring tools are configured.
Post-Condition	System issues can be detected and resolved efficiently.

7 Other non-functional attributes

Specifies any other particular non-functional attributes required by the system. Examples are provided below.

7.1 Security

7.2 Binary Compatibility

7.3 Reliability

7.4 Maintainability

7.5 Portability

7.6 Extensibility

7.7 Re-usability

7.8 Application Affinity/Compatibility

7.9 Resource Utilization

7.10 Serviceability

7.11 others as appropriate

8 Preliminary Object-Oriented Domain Analysis

This section identifies the fundamental objects that must be modeled within the InclusiveHire Platform in order to satisfy the system requirements. The purpose is to provide an alternative structural view of the system by describing how the requirements presented in earlier sections can be represented using object-oriented concepts.

The system is designed around several key domain objects that represent the major entities involved in inclusive hiring. These objects interact with one another to support the platform's core functionalities such as job posting, candidate applications, AI-based job matching, accessibility support, and system administration.

8.1 Inheritance Relationships

This section describes the primary inheritance hierarchy in the InclusiveHire Platform. Inheritance represents an "is-a" relationship, where a subclass inherits attributes and behaviors from a more general superclass.

In this system, the main superclass is User, which represents any individual who interacts with the platform. The User class contains common attributes such as user ID, name, email, password, and general system operations like login and profile management.

Three specialized classes inherit from the User class:

Table 21: NON - Other Requirements (Usability)

Attribute Name	Usability
Description	The system shall provide a user-friendly and intuitive interface requiring minimal training.
Critically	High – Directly impacts user satisfaction.
Technical issues	UI/UX design, accessibility standards, responsive layout.
Cost and schedule	Medium – Requires UI testing and improvements.
Risks	Poor interface design may reduce user adoption.
Dependencies with other requirements	Depends on frontend implementation and portability.
Pre-Condition	UI design guidelines are defined.
Post-Condition	Users can efficiently interact with the system.

Candidate – Represents job seekers with disabilities who use the platform to upload their CVs, receive job recommendations, and apply for jobs.

Employer – Represents organizations or HR representatives who post job opportunities, review candidate applications, and manage recruitment activities.

Administrator – Represents system managers responsible for maintaining the platform, managing users, monitoring job postings, and ensuring the system operates correctly.

This inheritance hierarchy allows the system to reuse shared user attributes and operations while allowing each subclass to define additional properties and behaviors specific to its role.

Conceptually, the inheritance structure can be described as:

User → Candidate → Employer → Administrator

This hierarchy improves system organization, reduces redundancy, and supports scalable object-oriented design within the InclusiveHire Platform.

8.2 Class descriptions

This section provides detailed descriptions of the main classes identified in the object-oriented domain analysis of the InclusiveHire Platform. Each class description follows a consistent structure to clarify the class role, its relationships, and how it supports system requirements.

8.2.1 Class name

8.2.2 List of Superclasses:

Names all immediate superclasses.

8.2.3 List of Subclasses:

Names all immediate subclasses.

8.2.4 Purpose:

States the basic purpose of the class.

8.2.5 Collaborations:

Names each class with which this class must interact in order to accomplish its purpose, and how.

8.2.6 Attributes:

Lists each attribute (state variable) associated with each instance of this class, and indicates examples of possible values (or a range).

8.2.7 Operations

: Lists each operation that can be invoked upon instances of this class. For each operation, the arguments (and their type), the return value (and its type), and any side effects of the operation should be specified.

8.2.8 Constraints:

Lists any restrictions upon the general state or behavior of instances of this class.

9 Operational Scenarios

9.1 Scenario 1: Candidate Registers and Creates a Profile

Actor: Candidate

Description: A new candidate creates an account, completes a disability profile, and uploads a CV. The system extracts skills and generates a structured profile.

1. The candidate opens the platform website.
2. The candidate selects the "Register" option.
3. The candidate enters personal information (name, email, password, phone).
4. The candidate completes the disability profile and ability information.
5. The candidate uploads a CV file.
6. The system performs OCR/NLP to extract skills and key information from the CV.
7. The system stores the candidate profile and confirms successful registration.

9.2 Scenario 2: Employer Posts a Job Opportunity

Actor: Employer

Description: An employer logs in, creates a job posting, and specifies required skills and accommodations.

1. The employer logs into the platform.
2. The employer navigates to the "Post Job" page.
3. The employer enters job details (title, description, location).
4. The employer specifies required skills and physical/environmental requirements.

5. The employer adds accessibility accommodations if applicable.
6. The system validates the job information.
7. The system publishes the job and makes it visible to candidates.

9.3 Scenario 3: Candidate Applies for a Job

Actor: Candidate

Description: A candidate reviews job recommendations, views the compatibility score, and submits an application.

1. The candidate logs into the platform.
2. The candidate browses recommended job opportunities.
3. The candidate selects a job to view its details.
4. The system displays job requirements and the candidate's compatibility score.
5. The candidate selects "Apply" for the job.
6. The system records the application and assigns an initial status (Submitted).
7. The system sends a notification to the employer.

9.4 Scenario 4: Employer Reviews Applications

Actor: Employer

Description: An employer reviews applicants ranked by compatibility score and updates application decisions.

1. The employer logs into the platform.
2. The employer opens the job posting dashboard.
3. The system displays the list of applications for the selected job.
4. The employer reviews candidate profiles and compatibility scores.
5. The employer selects an application and chooses Accept or Reject.
6. The system updates the application status accordingly.
7. The system sends notifications to the candidate about the decision.

10 Preliminary Schedule Adjusted

10.1 Task 1: Requirements Analysis

- Duration: 2 weeks
- Output: SRS document, initial requirements validation

10.2 Task 2: System Design

- Duration: 2 weeks
- Output: Architecture design, database schema, UML diagrams

10.3 Task 3: Frontend Development

- Duration: 3 weeks
- Output: UI implementation, accessibility features, user flows

10.4 Task 4: Backend Development

- Duration: 3 weeks
- Output: APIs, authentication, database operations

10.5 Task 5: AI Integration

- Duration: 2 weeks
- Output: CV parsing, skill extraction, compatibility scoring, recommendations

10.6 Task 6: Testing and Debugging

- Duration: 2 weeks
- Output: Functional tests, performance tests, usability validation

10.7 Task 7: Deployment

- Duration: 1 week
- Output: Cloud deployment, monitoring setup, final validation

10.8 Overall Timeline Summary

- Total Estimated Duration: 14 weeks
- Notes: Task dependencies should be represented using a Gantt or PERT chart.

11 Preliminary Budget Adjusted

11.1 Cost Factor 1: Development Costs

- Description: Expenses related to implementation and design efforts.
- Estimated Share: 50% of total budget

11.2 Cost Factor 2: Cloud Hosting Services

- Description: Server hosting, database storage, and network services.
- Estimated Share: 20% of total budget

11.3 Cost Factor 3: AI Processing Services

- Description: OCR/NLP processing tools, model computation resources.
- Estimated Share: 15% of total budget

11.4 Cost Factor 4: Testing and Maintenance

- Description: Testing tools, debugging time, and maintenance planning.
- Estimated Share: 10% of total budget

11.5 Cost Factor 5: Miscellaneous Costs

- Description: Licenses, third-party integrations, operational needs.
- Estimated Share: 5% of total budget

12 Appendices

This section contains additional information that helps in understanding the requirements and terminology used in the Software Requirements Specification (SRS).

12.1 Definitions, Acronyms, Abbreviations

- **AI (Artificial Intelligence)** – Computer systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, and decision-making.
- **OCR (Optical Character Recognition)** – A technology used to convert scanned documents or images of text into machine-readable text.
- **NLP (Natural Language Processing)** – A field of artificial intelligence that focuses on enabling computers to understand and process human language.
- **SRS (Software Requirements Specification)** – A document that describes the intended purpose, features, and constraints of a software system.
- **WCAG (Web Content Accessibility Guidelines)** – International standards that provide recommendations for making web content accessible to people with disabilities.
- **CV (Curriculum Vitae)** – A document that summarizes a candidate's education, skills, and work experience.

- **Compatibility Score** – A numerical value (0–100) generated by the system to indicate how well a candidate matches a specific job.
- **Accommodation** – Adjustments or modifications in the workplace that support employees with disabilities.

12.2 Collected material

This appendix contains supporting materials used during the preparation of this SRS document.

- Research articles and reports related to inclusive hiring and employment for people with disabilities.
- Documentation related to accessibility standards such as WCAG.
- Technical documentation for AI tools used in the system, including OCR and NLP libraries.
- Reference materials on software engineering methodologies and requirements analysis.
- Background information gathered from existing job recruitment platforms.

13 References

References